MATH 201 Your name: _____

**Learning target 2.3, version 1**

Compute derivatives using the product and quotient rules.

---

1. Consider the function $g(t) = 7 \cdot t^4$.

   (a) Compute $g'(t)$ using the product rule. Simplify as much as possible.

   (b) Now compute $g'(t)$ again, *without* the product rule.

   (c) Are both ways *correct*? Which way do you *prefer*?

2. Consider the function $f(t) = \dfrac{t^4}{7}$.

   (a) Compute $f'(t)$ using the quotient rule. Simplify as much as possible.

   (b) Now compute $f'(t)$ again, *without* the quotient rule.
   (Hint: What's another way to say "divide by 7"?)

   (c) Are both ways *correct*? Which way do you *prefer*?

MATH 201 Your name: _____

## Learning target 2.3, version 2

Compute derivatives using the product and quotient rules.

---

Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (product, quotient, sum and difference, etc.) you are using.

1. $h(w) = -\dfrac{\cos(w)}{6w^2+5w+4}$

2. $g(w) = \dfrac{3w^2+5w-1}{w^8}$

3. $f(w) = -\left(4w^2+6w+1\right)\ln(w)$

MATH 201 Your name: _____

**Learning target 2.3, version 3**

Compute derivatives using the product and quotient rules.

---

Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (product, quotient, sum and difference, etc.) you are using.

1. $g(w) = -\dfrac{2w^2+6w+1}{w^{1/5}}$

2. $f(w) = -\dfrac{\sin(w)}{3w^2-3w+4}$

3. $h(w) = -\left(2w^2+5w-5\right)e^w$

MATH 201 Your name: _____

**Learning target 2.3, version 4**

Compute derivatives using the product and quotient rules.

---

Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (product, quotient, sum and difference, etc.) you are using.

1. $f(x) = -\left(3x^2 - 6x + 2\right)\cos(x)$

2. $h(y) = \dfrac{3y^2 - 3y - 2}{\sin(y)}$

3. $g(w) = \dfrac{2w^2 - w + 4}{w^{1/6}}$

MATH 201 Your name: _____

**Learning target 2.3, version 5**

Compute derivatives using the product and quotient rules.

---

Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (product, quotient, sum and difference, etc.) you are using.

1. $g(x) = \dfrac{\ln(x)}{2x^2+6x-3}$

2. $f(x) = \left(x^2-2x-4\right)\cos(x)$

3. $h(t) = -\dfrac{3t^2+2t+3}{t^3}$